Hibiscus anthocyanins rich extract-induced apoptotic cell death in human promyelocytic leukemia cells.

Hibiscus sabdariffa Linne (Malvaceae), an attractive plant believed to be native to Africa, is cultivated in the Sudan and Eastern Taiwan. Anthocyanins exist widely in many vegetables and fruits. Some reports demonstrated that anthocyanins extracted from H. sabdariffa L., Hibiscus anthocyanins (HAs) (which are a group of natural pigments existing in the dried calyx of H. sabdariffa L.) exhibited antioxidant activity and liver protection. Therefore, in this study, we explored the effect of HAs on human cancer cells. The result showed that HAs could cause cancer cell apoptosis, especially in HL-60 cells. Using flow cytometry, we found that HAs treatment (0-4 mg/ml) markedly induced apoptosis in HL-60 cells in a dose- and time-dependent manner. The result also revealed increased phosphorylation in p38 and c-Jun, cytochrome c release, and expression of tBid, Fas, and FasL in the HAs-treated HL-60 cells. We further used SB203580 (p38 inhibitor), PD98059 (MEK inhibitor), SP600125 (JNK inhibitor), and wortmannin (phosphatidylinositol 3-kinase; PI-3K inhibitor) to evaluate their effect on the HAs-induced HL-60 death. The data showed that only SB203580 had strong potential in inhibiting HL-60 cell apoptosis and related protein expression and phosphorylation. Therefore, we suggested that HAs mediated HL-60 apoptosis via the p38-FasL and Bid pathway. According to these results, HAs could be developed as chemopreventive agents. However, further investigations into the specificity and mechanism(s) of HAs are needed.